Models and Tools for Cyber-Physical Systems

Chapter 7: Dynamical Systems – Part 1

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Original slides courtesy of Rajeev Alur
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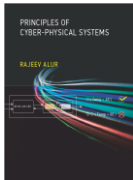
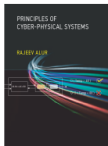


Dynamical Systems

- Many different mathematical representations and definitions: https://en.wikipedia.org/wiki/Dynamical_system

- The underlying intention of these definitions is to model the evolution of a system's states over time.

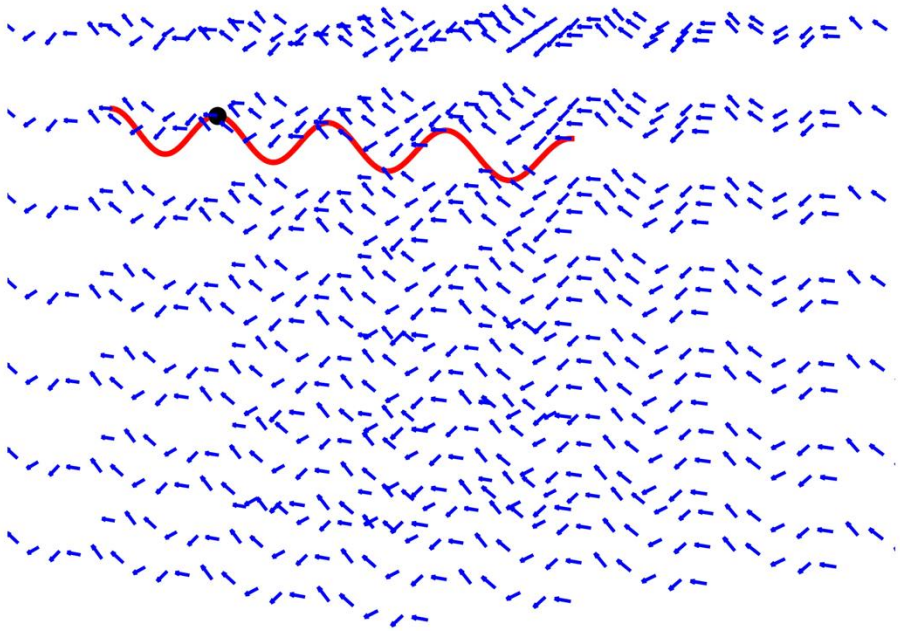
- This course: if T represents time and S represents a system's state space, a "dynamical system" is a function $\varphi: T \to S \to S$ (alt. $\varphi: U \to S$, where $\mathrm{U} \subseteq T \times S$) such that:
  - $\varphi(0)(s) = s$ (at time 0 any state is mapped to itself)
  - $\varphi(t_1 + t_2) = \varphi(t_1) \circ \varphi(t_2)$ (determinism of the system)



Dynamical Systems (formal) examples

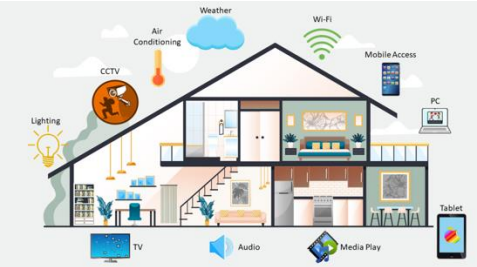
- A clock as a (discrete) dynamical system
  - $\varphi\,(0, 11{:}34) = 11{:}34$ $\quad\varphi\,(1, 11{:}34) = 11{:}35$ $\quad\varphi\,(2, 11{:}34) = 11{:}36$
    $\varphi\,(60, 11{:}34) = 0{:}34$ $\quad\varphi\,(480, 11{:}34) = 11{:}34$ $\quad\varphi\,(0, 11{:}35) = 11{:}35$
- A continuous dynamical system

$$\varphi\,(t)(x, y, z) = (x + v \cdot t, y, z - \cos x + \cos(x + v \cdot t))$$



*check that the equations $\varphi(0)(s) = s$ and $\varphi(t_1 + t_2) = \varphi(t_1) \circ \varphi(t_2)$ hold

Dynamical Systems modeling examples



Dynamical Systems

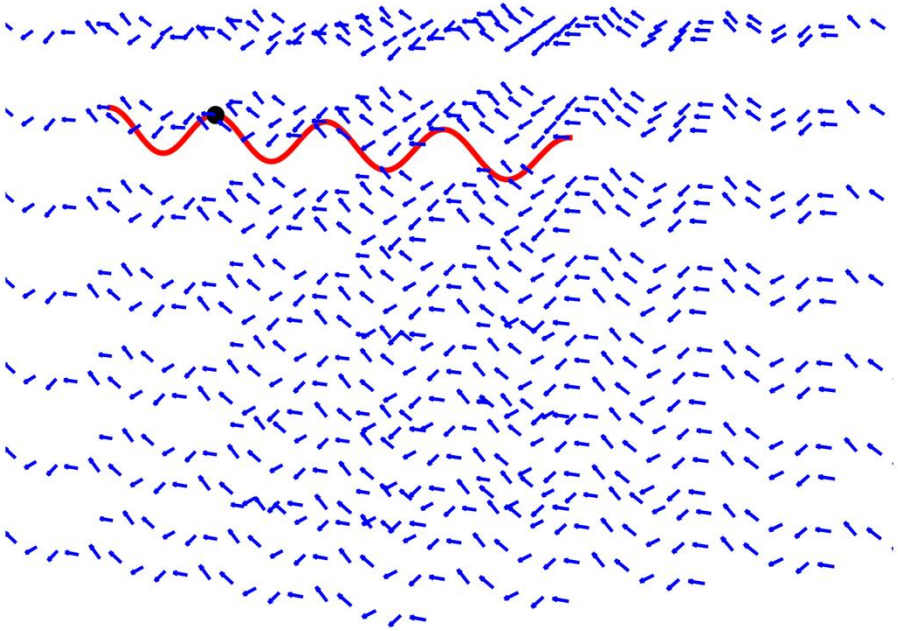
- Common approach to model continuous dynamical systems: ordinary differential equations (ODEs)

$$x'(t) = v \qquad y'(t) = 0 \qquad z'(t) = -\sin\big(x(t)\big)$$

  - "The particle moves constantly in one direction"
  - "The particle does not move in the orthogonal direction"
  - "The particle oscillates vertically as it moves:

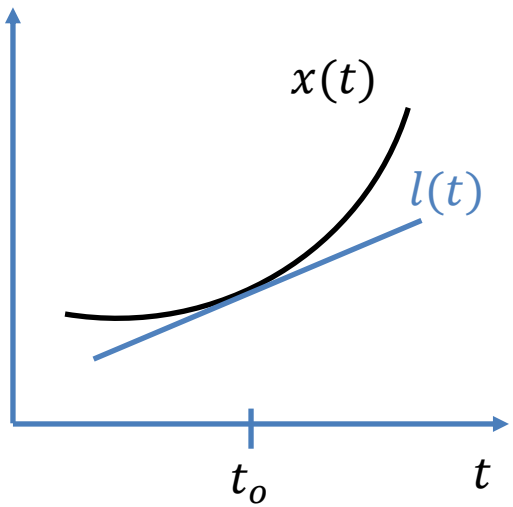
- Variables: Physical quantities evolving continuously over time
  - Temperature, pressure, velocity …

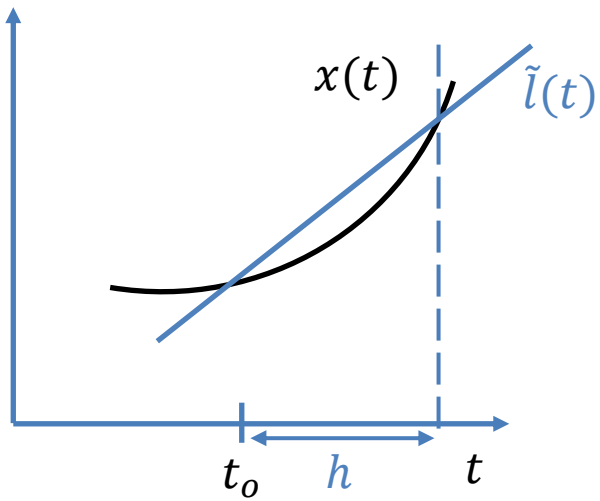
- Governed by nature's laws
  - Newton's laws
  - Laws of thermodynamics
  - Maxwell's equations



Calculus Brush Up

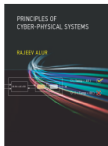
- $\frac{d}{dt}x(t)$ captures the rate of change of variable x at time t, i.e. the slope
- $\frac{d}{dt}x(t)$ is defined as the limit of the difference: $\frac{d}{dt}x(t) = \lim_{h \to 0} \frac{x(t+h)-x(t)}{h}$





$$l(t) = x(t_0) + \frac{d}{dt}x(t_0)(t - t_0) \qquad l(t) \approx \tilde{l}(t) = x(t_0) + \left(\frac{x(t_0 + h) - x(t_0)}{h}\right)(t - t_0)$$

* Notations for derivatives: $\frac{\mathrm{d}}{\mathrm{dt}}\mathrm{x(t)}$ or $\mathrm{x'(t)}$ or $\dot{x}(\mathrm{t})$



What is an ODE?

Relation between a function $z : \mathbb{R} \to \mathbb{R}$ and its derivatives

$$F\left(z\,t, \dot{z}\,t, \ddot{z}\,t, \ldots, z^{(k)}\,t\right) = 0$$

rewritten by the implicit function theorem

kth-order ODE

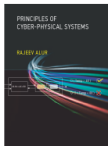
$$z^{(k)} = g\left(t, z\,t, \dot{z}\,t, \ddot{z}\,t, \ldots, z^{(k-1)}\,t\right)$$

renaming $x_1\,t \triangleq z\,t$ and $x_i\,t \triangleq \dot{x}_{i-1} = z^{(i-1)}$

Important trick (for later)!

k 1st-order ODEs

$$\begin{aligned}
\dot{x}_1\,t &= x_2\,t \\
\dot{x}_2\,t &= x_3\,t \\
&\vdots \\
\dot{x}_{k-1}\,t &= x_k\,t \\
\dot{x}_k\,t &= g\left(t, x_1\,t, x_2\,t, x_3\,t, \ldots, x_k\,t\right)
\end{aligned}$$



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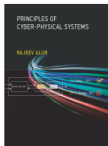
and $x_i\,t \triangleq \dot{x}_{i-1} = z^{(i-1)}$

$x_0\,t \triangleq t$

k+1
1st-order ODEs

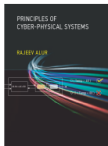
$$\begin{aligned}
\dot{x}_0\,t &= 1 \\
\dot{x}_1\,t &= x_2\,t \\
\dot{x}_2\,t &= x_3\,t \\
&\vdots \\
\dot{x}_{k-1}\,t &= x_k\,t \\
\dot{x}_k\,t &= g\left(x_0\,t, x_1\,t, x_2\,t, x_3\,t, \ldots, x_k\,t\right)
\end{aligned}$$



What is an ODE?

Focus then on (autonomous) systems of n first-order ODEs

$$
\begin{aligned}
\dot{x}_1\, t &= f_1\,(x_1\, t, x_2\, t, x_3\, t, \ldots, x_n\, t) \\
\dot{x}_2\, t &= f_2\,(x_1\, t, x_2\, t, x_3\, t, \ldots, x_n\, t) \\
&\vdots \\
\dot{x}_n\, t &= f_n\,(x_1\, t, x_2\, t, x_3\, t, \ldots, x_n\, t)
\end{aligned}
$$



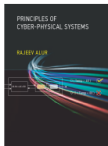
What is an ODE?

Focus then on (autonomous) systems of n first-order ODEs

$$\dot{X}\,t = \begin{pmatrix} \dot{x}_1\,t \\ \dot{x}_2\,t \\ \vdots \\ \dot{x}_n\,t \end{pmatrix} = \begin{pmatrix} f_1\,(x_1\,t, x_2\,t, x_3\,t, \ldots, x_n\,t) \\ f_2\,(x_1\,t, x_2\,t, x_3\,t, \ldots, x_n\,t) \\ \vdots \\ f_n\,(x_1\,t, x_2\,t, x_3\,t, \ldots, x_n\,t) \end{pmatrix} = f\,(X\,t)$$

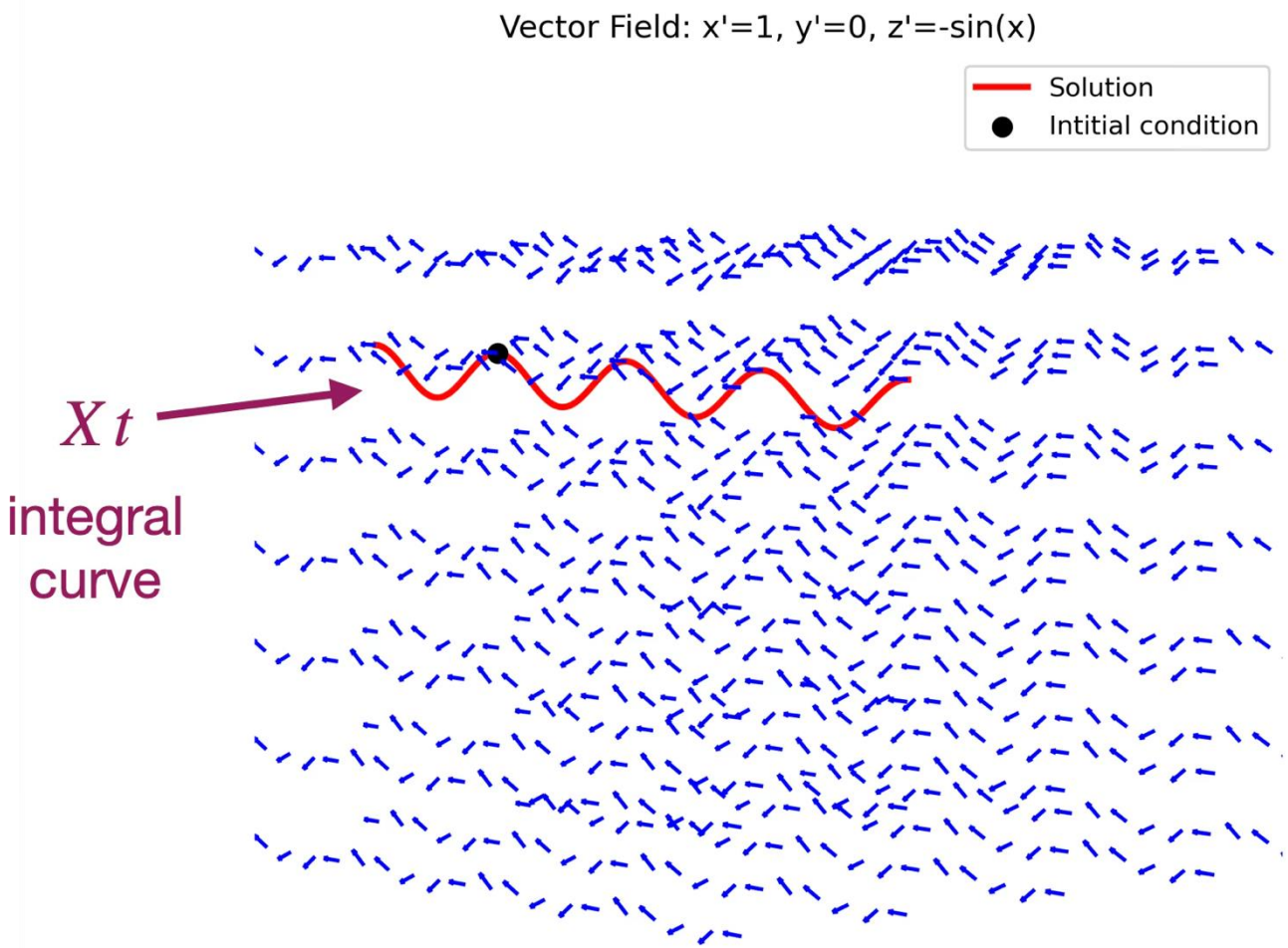
$X : \mathbb{R} \to \mathbb{R}^n$ is a *solution/integral curve*

$f : \mathbb{R}^n \to \mathbb{R}^n$ is a *vector field*



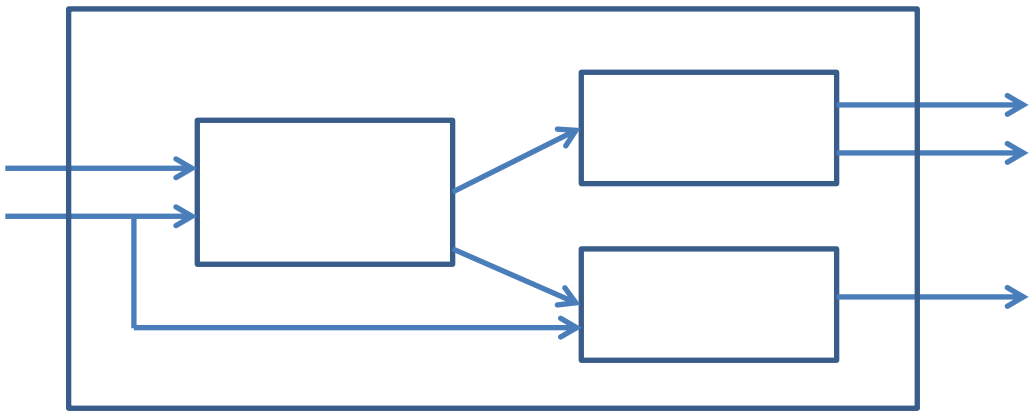
What is an ODE?

ODEs are vector fields!



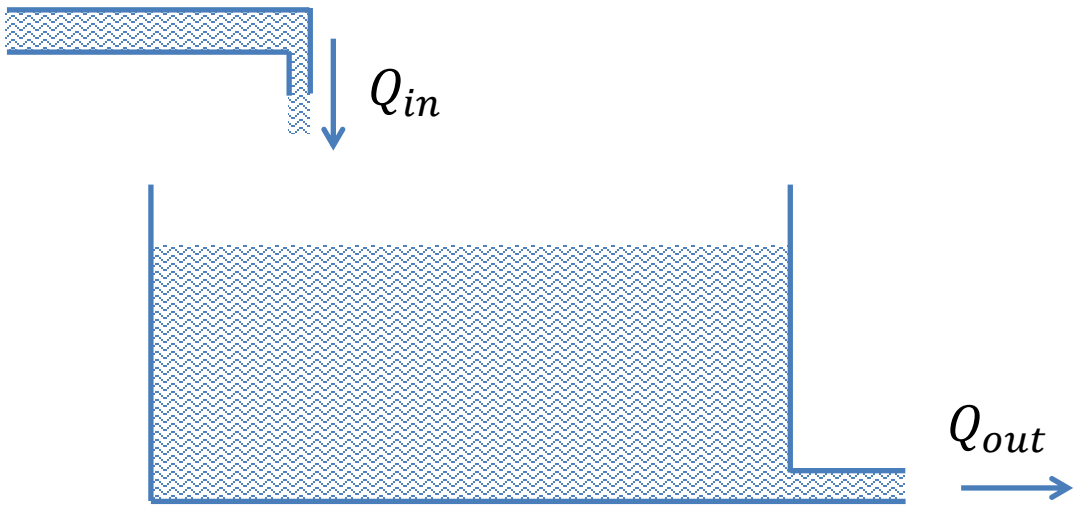
Initial value problem: $\dot{x}(\mathrm{t}) = f\big(x(t)\big)$ with $\mathrm{x}(0) = \mathrm{x}_0$

Model-Based Design



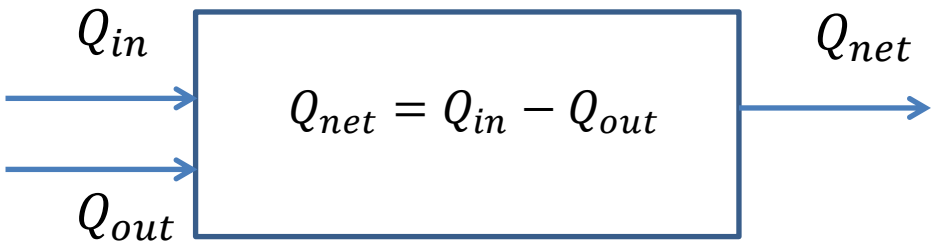
- Block Diagrams
  - Widely used in industrial design
  - Tools: Simulink, Modelica, RationalRose…
- Key question: what is the execution semantics?
  - Similar to **synchronous** model, but continuous-time instead of discrete-time

Example: Water Flow in a Tank



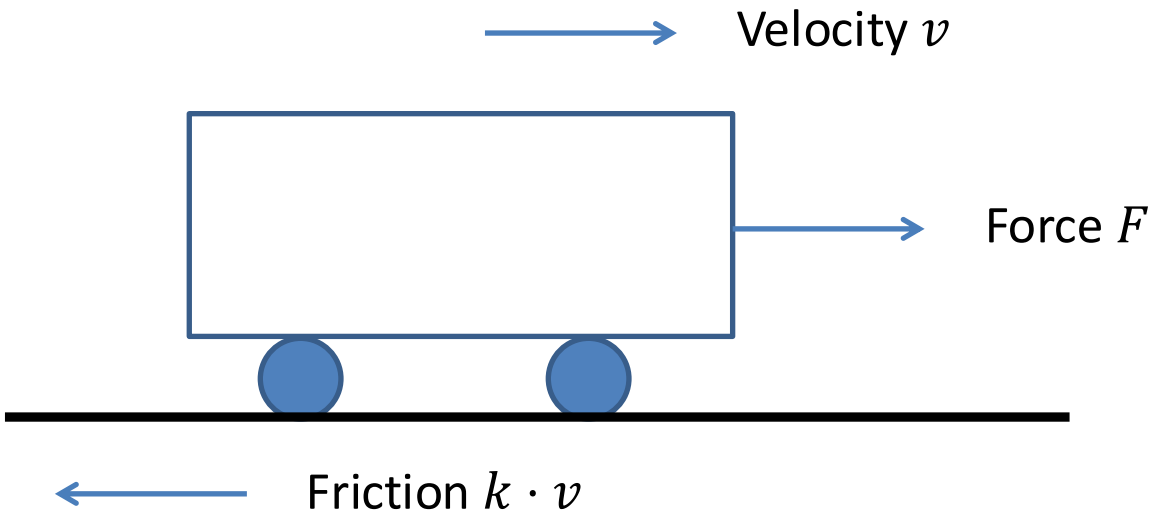
- Inflow of water Q_{in}
- Outflow of water Q_{out}

Example: Water Flow in a Tank



- Input variables: Q_{in} and Q_{out} of type real
- Output variable: Q_{net} of type real
- No state variables
- Signal: assignment of values to variables as function of time t
- At each time t, value of output signal $Q_{net}(t)$ equals $Q_{in}(t) - Q_{out}(t)$
- Output as a function of inputs/state specified using algebraic equations (as opposed to assignments)

Example: Car Model

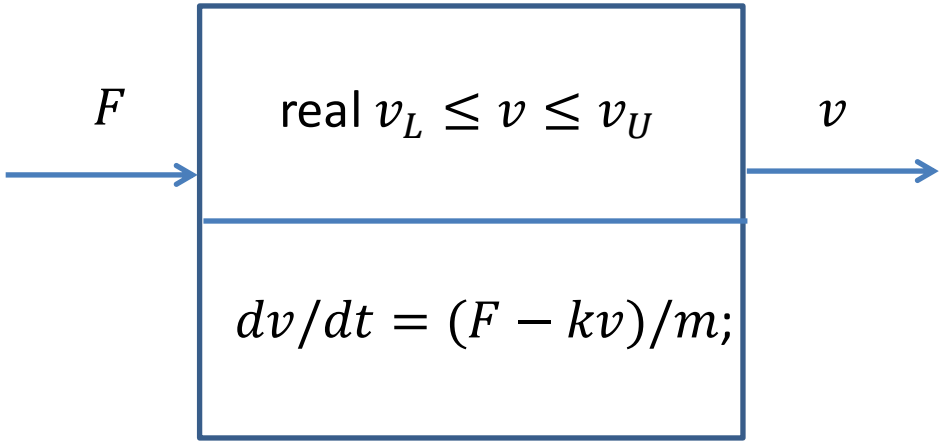


Newton's laws of motion:

1. Bodies remain at rest, or in motion at a constant speed (a= 0) in a straight line ($\sum F = 0$), unless acted upon by a force ($\sum F \neq 0$).
2. $F_{net} = m \cdot a$
3. If two bodies exert forces on each other, these forces have the same magnitude but opposite directions

$$F_{net} = m \cdot a \rightarrow F - kv = m \cdot v'$$

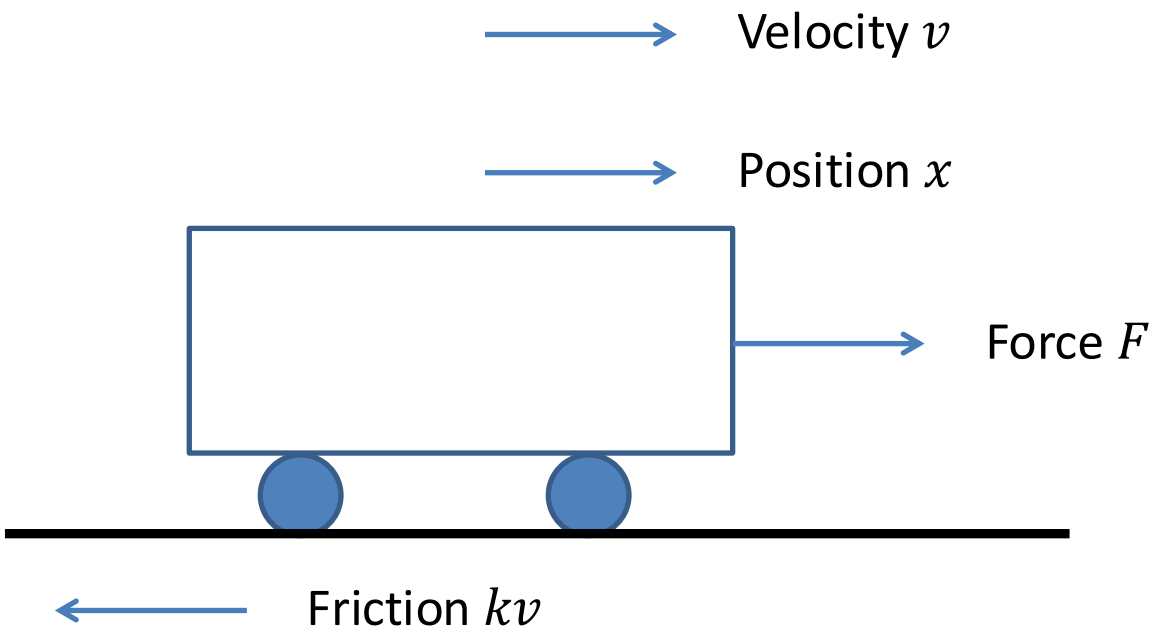
Continuous-time Component Car



- For state variable v, its rate of change dv/dt is defined using an expression over input and state variables

- For each output variable, its value is defined using an expression over input and state variables.

Example: Car Model with two variables



Newton's law of motion gives

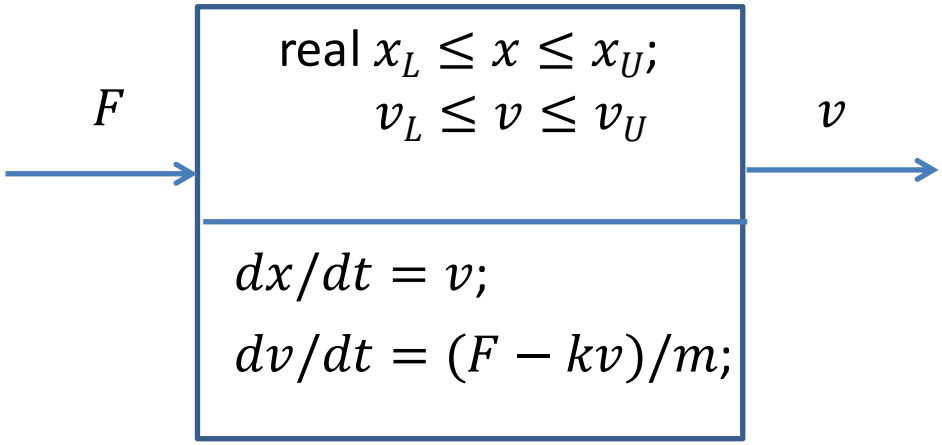
$$F_{net} = ma \rightarrow$$

$$F - kv = m\, d^2x/dt$$

Continuous-time Component Car

- Remember that $a = dv/dt$ and $v = dx/dt$

- We can rewrite $F - kv = m\, d^2x/dt$ into the system below using state variables x and v.



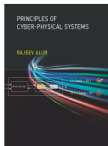
Executions of Car

- Input signal: Function $\boldsymbol{F}(t) : real_{\geq 0} \rightarrow real$ that gives value of force as a function of time
  - Should be continuous or piecewise-continuous

- Given an initial state (x_0, v_0) and input signal $\boldsymbol{F}(t)$, the execution of the system is defined by state-signals $\boldsymbol{x}(t)$ and $\boldsymbol{v}(t)$ (from $real_{\geq 0}$ to $real$) that satisfy the initial-value problem:
  - $\boldsymbol{x}(0) = x_0$; $\boldsymbol{v}(0) = v_0$
  - $d\boldsymbol{x}(t)/dt = \boldsymbol{v}(t)$
  - $d\boldsymbol{v}(t)/dt = (\boldsymbol{F}(t) - k\boldsymbol{v}(t)) \,/\, m$

Definition: Continuous-Time Component

- Set I of real-valued input variables
  - Type is either real or interval of real, real$[L, U]$
- Set O of real-valued output variables
- Set S of real-valued state variables
- Initialization $Init$ specifying set $[Init]$ of initial states
- For each output var y, a real-valued expression h_y over $I \cup S$
- For each state variable x, a real-valued expression f_x over $I \cup S$
- Given an input-signal $\boldsymbol{I}(t)$: real$_{>=0}$ -> real$^{|I|}$, an execution consists of a *differentiable* state signal $\boldsymbol{S}(t)$ and output signal $\boldsymbol{O}(t)$ such that
  1. $\boldsymbol{S}(0)$ is in $[Init]$
  2. For each output var y and time t, $\boldsymbol{y}(t) = hy(\boldsymbol{I}(t), \boldsymbol{S}(t))$
  3. For each state var x, $(d/dt)\ (\boldsymbol{x}(t)) = fx(\boldsymbol{I}(t), \boldsymbol{S}(t))$



Theorem of the existence and uniqueness of solutions to IVPs

PICARD–LINDELÖF THEOREM

Existence and Uniqueness

- For the initial value problem $x' = if\ x = 0\ then\ 1\ else\ 0$ (on an interval around 0) and $x(0) = 0$:
  - Suppose there is a solution x to that IVP
  - Since $x(0) = 0$, the derivative satisfies $x'(0) = 1$
  - Since the derivative is a positive limit, we can find a value t near 0, where $x(t) \neq 0$
  - Hence, $x'(t) = 0$, and by Darboux's theorem, for each number k between 0 and 1 there must be a $c \in (0, t)$ such that $x'(c) = k$. A contradiction!
  - Therefore, there does **<u>not exist</u>** solution that solves that IVP

- For the initial value problem $y' = y^{1/3}$ and $y(t_0) = 0$:
  - By inspection $y(t) = 0$ solves the IVP
  - But also $\int dy/y^{1/3} = \int t = t + C \quad \Rightarrow \quad \frac{3}{2} y^{\frac{2}{3}} = t + C \quad \Rightarrow \quad y(t) = \pm\left(\frac{2}{3} \cdot (t + C)\right)^{3/2}$
  - Thus $y(t) = \pm\left(\frac{2}{3} \cdot (t - t_0)\right)^{3/2}$ and we have found **<u>more than one</u>** solution to the IVP

Existence and Uniqueness

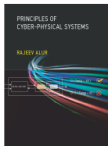
- Given an input signal $\boldsymbol{I}(t)$, when are we guaranteed that the system has at least one execution? Exactly one execution?

- The input signal should be continuous (or at least piecewise continuous), but also depends on right-hand-sides of equations defining state and output dynamics

- Consider the initial value problem

  $dx/dt = f(x);\ x(0) = x_0$, x is k-dimensional signal

- When does there exist a unique differentiable function $\boldsymbol{x}(t)$ as a solution?

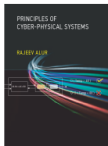


Existence

- Consider the initial value problem

  $dx/dt = f(x)$; $x(0) = x_0$, x is k-dimensional signal
- There exists at least one solution $\boldsymbol{x}(t)$ if the function f is a continuous function (Peano existence theorem)
- Definition of continuity relies on definition of distance between points (e.g. Euclidean distance)
- The function f is (uniformly) continuous if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all u, v in real^k, if $\|u - v\| < \delta$ then $\|f(u) - f(v)\| < \varepsilon$
- Example when solution does not exist:

  dx/dt = if $(x = 0)$ then 1 else 0
- Natural to require all right-hand-side expressions h_y and f_x in definition of a continuous-time component to be continuous
  - Discontinuous case -> Hybrid Systems



Uniqueness

- Consider the initial value problem

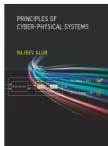
  $dx/dt = f(x);\ x(0) = x_0$, x is k-dimensional signal

- Picard-Lindelöf Theorem: There exists a unique solution $\boldsymbol{x}(t)$ if the function f is Lipschitz-continuous

- Informally, Lipschitz-continuous means there is a constant upper bound on how fast a function changes

- Function f is Lipschitz-continuous if there exists a constant K such that for all u, v in real^k, $\|f(u) - f(v)\| \leq K\|u - v\|$

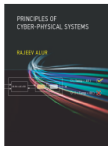
- Theorem: if f is continuously differentiable (has a derivative and the derivative is continuous), then f is (locally) Lipschitz-continuous



Uniqueness

- Examples:
  - A linear function such as $(F - kv)/m$ is Lipschitz-continuous
  - Quadratic function x^2 ? Lipschitz-cont if domain of x is bounded

- $x^{1/3}$ is not Lipschitz-cont ($x \to 0$): $dx/dt = x^{1/3}$; $x(0) = 0$ has multiple solutions
  - $\boldsymbol{x}(t) = 0$
  - $\boldsymbol{x}(t) = (2t/3)^{3/2}$



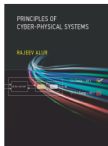
Lipschitz Continuous Component

- A continuous-time component has Lipschitz-continuous dynamics if
  - Each expression h_y corresponding to output variable y is a Lipschitz-continuous function of $I \cup S$
  - Each expression f_x corresponding to state variable x is a Lipschitz-continuous function over $I \cup S$

- If we supply a Lipschitz continuous component with a continuous input signal $\boldsymbol{I}(t)$, then it has a unique response signals $\boldsymbol{S}(t)$ and $\boldsymbol{O}(t)$, both of which are continuous

- Note: Continuity of output signals means these can be fed to other components in a block diagram

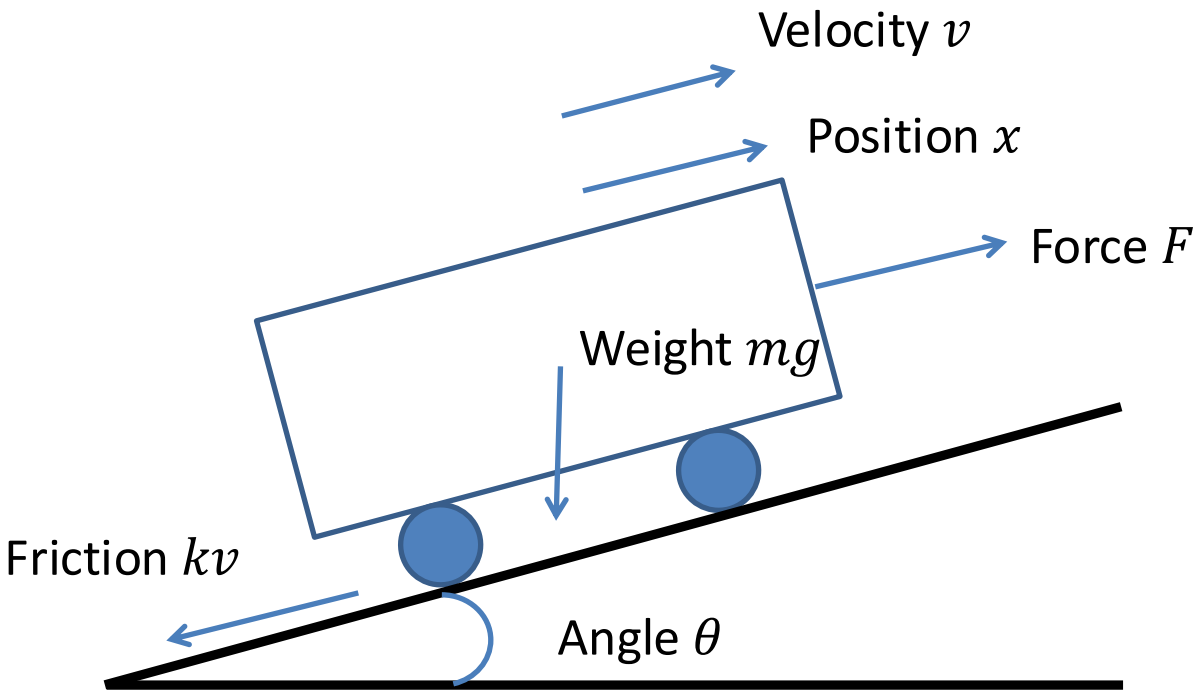
- Henceforth, we will assume all components are Lipschitz continuous



Models with disturbance

STABILITY

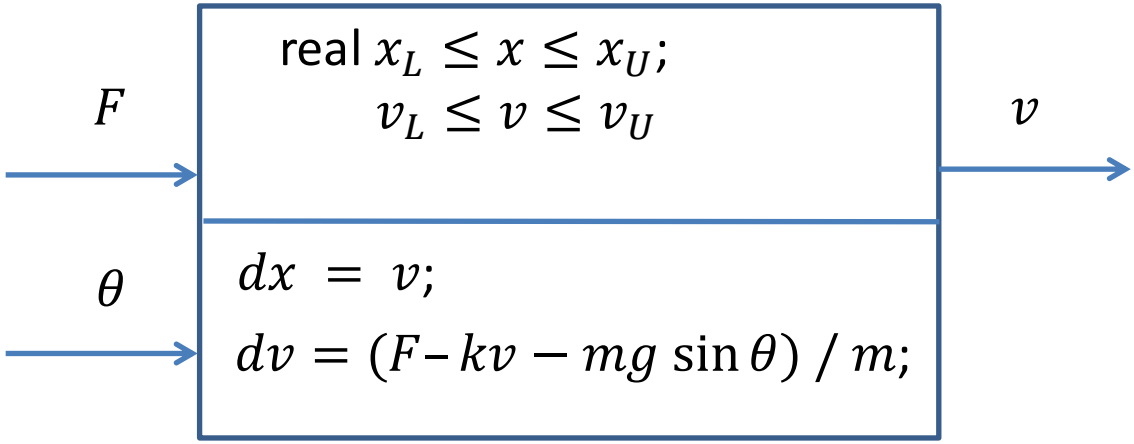
Car on a graded road – uncontrollable unputs



Newton's law of motion gives

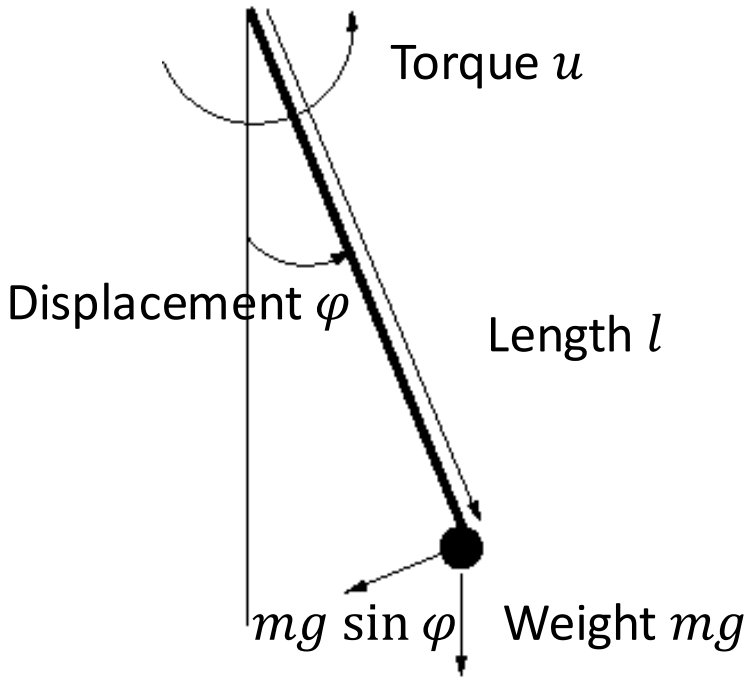
$$F - kv - mg \sin\theta = m\, d^2x/dt$$

Continuous-time Component Car2



- The grade of the road, denoted θ, models disturbance or an uncontrolled input

Simple Pendulum – nonlinear dynamics



- External torque applied by the motor at the pivot: u
- Dynamics captured by the second-order non-linear differential eqn:

$$ml^2 \, (d^2/dt)\varphi = u - mgl \sin \varphi$$

Pendulum Model

- Dynamics captured by the second-order non-linear differential eqn:

$$ml^2\,(d^2/dt)\varphi = u - mgl\,\sin\varphi$$

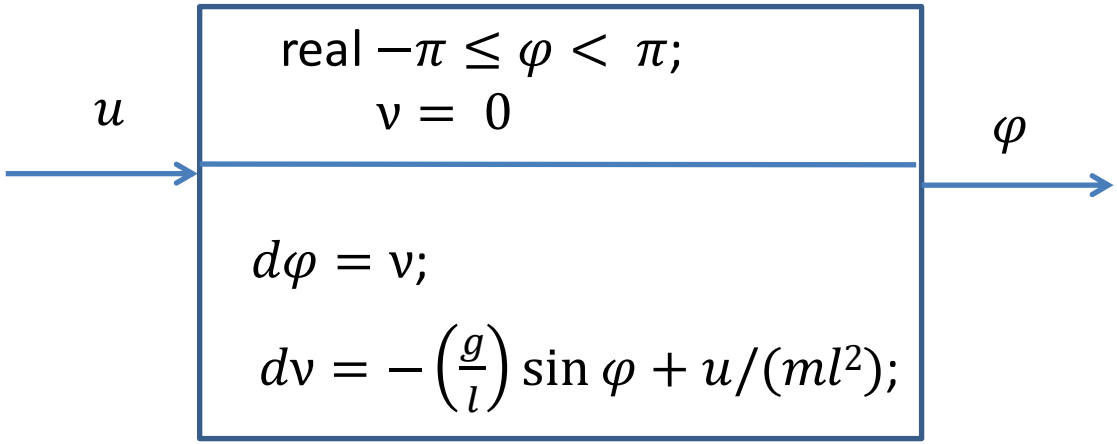
- Note that $\frac{d}{dt}\varphi = \nu$, so $\frac{d^2}{dt}\varphi = \frac{d}{dt}\left(\frac{d}{dt}\varphi\right) = \frac{d}{dt}\nu$

- With rewriting

$$ml^2\left(\frac{d^2}{dt}\right)\varphi = u - mgl\,\sin\varphi$$

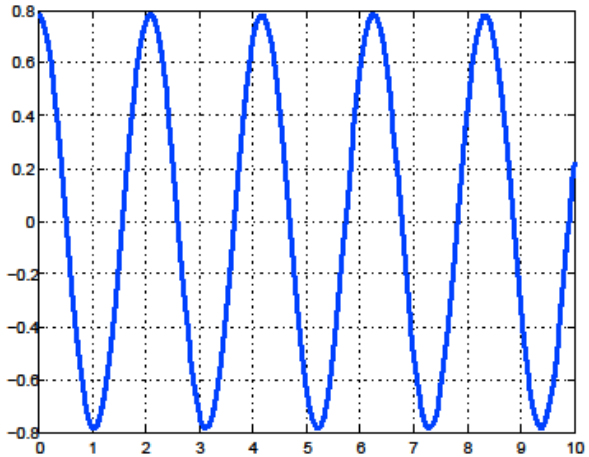
$$ml^2\frac{d}{dt}\nu = u - mgl\,\sin\varphi$$

$$\frac{d}{dt}\nu = \frac{u}{ml^2} - \frac{g}{l}\sin\varphi$$



Angular Displacement

- External torque = 0; Initial displacement = $\pi/4$
- Oscillatory motion plotted
- Consider $dx/dt = f(x)$. An x satisfying $0 = f(x)$ is called equilibrium
- What are equilibria of this pendulum ?



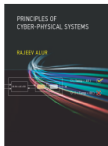
Equilibria of Dynamical Systems

- Consider a closed (i.e. without inputs) continuous-time component H
  - If H has inputs, then we can analyze equilibria by setting inputs to a fixed value
  - Assume state x is k-dimensional, and dynamics is Lipschitz-continuous given by $dx/dt = f(x)$

- A state x_e is called an equilibrium of H if $f(x_e) = 0$

- If $x'(t) = 0$, then x does not change over time. In control, this is phrased as "steady-state x".

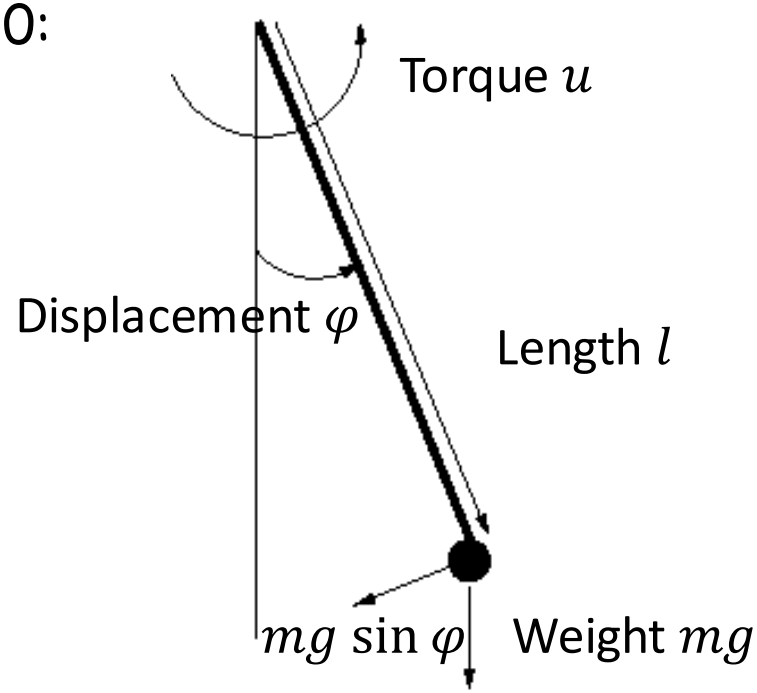
- If initial state of H equals an equilibrium state x_e, then the system stays in this state at all times



Pendulum Equilibria

Dynamics when external torque is 0:

$$d\varphi = \nu;\ d\nu = -\left(\frac{g}{l}\right)\sin\varphi$$



Equilibrium states:

$$\nu = 0;\ \sin\varphi = 0$$

Equilibrium state 1: $\nu = 0$; $\varphi = 0$; Pendulum is vertically downwards

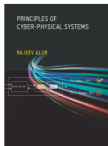
Equilibrium state 2: $\nu = 0$; $\varphi = -\pi$; Pendulum is vertically upwards

Stability of Dynamical Systems

- Key correctness requirement for dynamical systems: stability
  - Small perturbations in the input values should not cause disproportionately large changes in the outputs

- For cruise controller, correctness requirements:
  - Stability: If grade of the road changes, speed should change only slowly
  - Safety: Speed should always be within certain threshold values
  - Liveness: Actual speed should eventually get close to desired speed

- Classical mathematical formalization of stability:
  - Lyapunov stability of equilibria



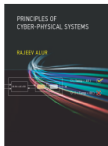
Lyapunov Stability

- Consider a closed continuous-time component H with Lipschitz-continuous dynamics $dx/dt = f(x)$

- Given an initial state s, let $\boldsymbol{x}[s]$ denote the unique state response signal for the initial value problem $x(0) = s$ and $dx/dt = f(x)$

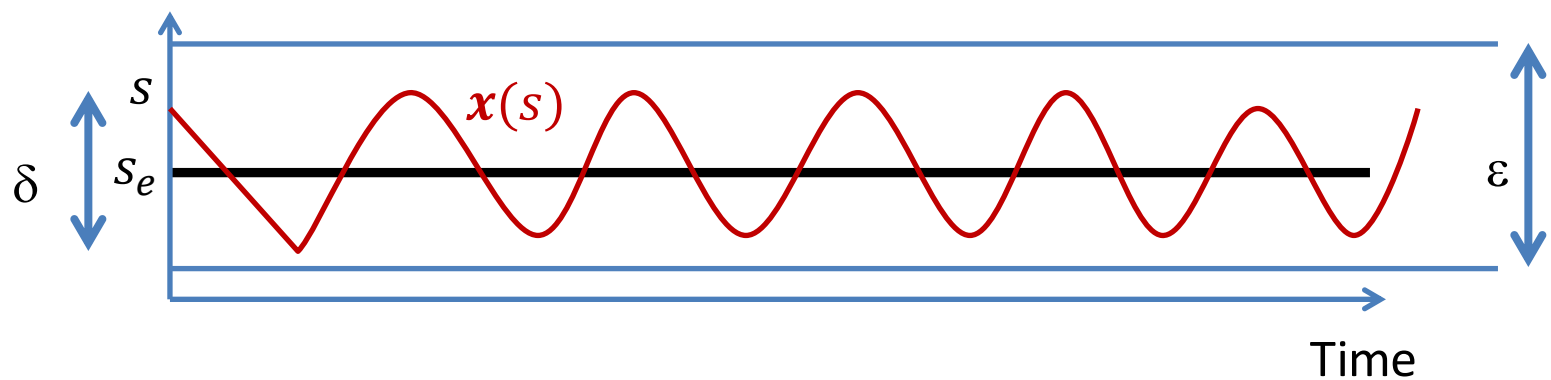
- Consider an equilibrium state s_e: if initial state is s_e then the response $\boldsymbol{x}[s_e]$ is a constant function of time, always equal to s_e

- Stability of an equilibrium: when the system is in an equilibrium state, if we perturb its state slightly
  - As time passes, will the state stay close to the equilibrium state ?
  - As time passes, will the system eventually return to the equilibrium state?



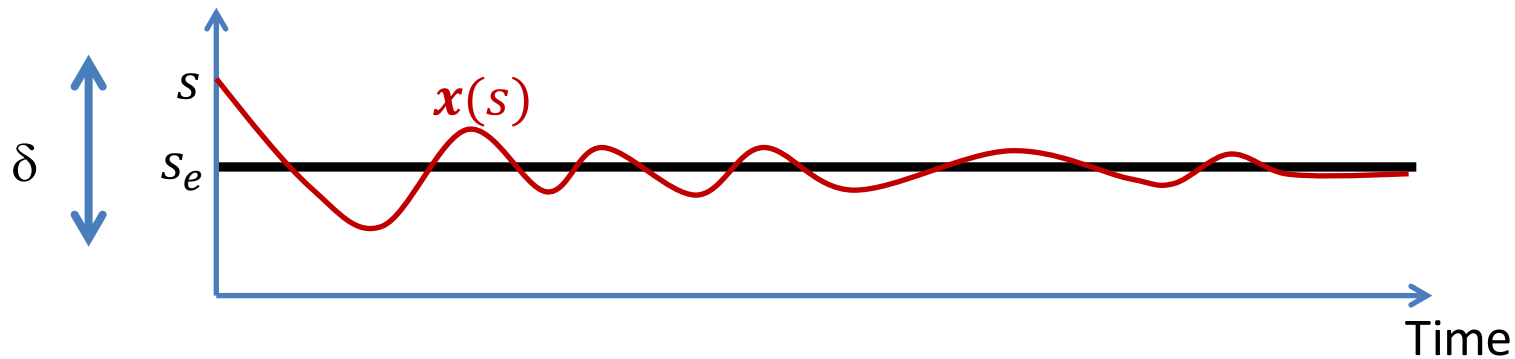
Lyapunov Stability Conditions

- Suppose the initial state s is *close* to an equilibrium state s_e, does the state along the response signal $\boldsymbol{x}[s]$ stay close to s_e?
  - If so, the equilibrium s_e is said to be stable
  - Formally, for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all initial states s if $\|s - s_e\| < \delta$, for all times $t \geq 0$, $\|x[s](t) - s_e\| < \varepsilon$



Lyapunov Stability Conditions

- If in addition, the response signal $x[s]$ converges to the equilibrium state s_e, then the equilibrium is asymptotically stable
  - There exists $\delta > 0$ such that for all states s if $\|s - s_e\| < \delta$ then the limit $\lim_{t \to \infty} x[s](t)$ exists and equals s_e
  - Note: This is a stronger condition than stable



http://youtu.be/W8YpgG0KuOo

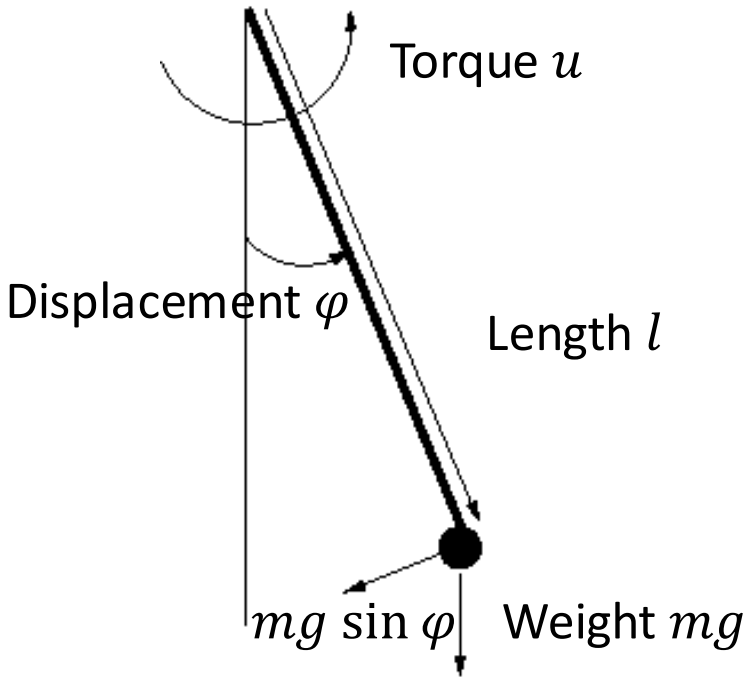
Pendulum Equilibria

Equilibrium state 1: $\nu = 0$; $\varphi = 0$;
Pendulum is vertically downwards.

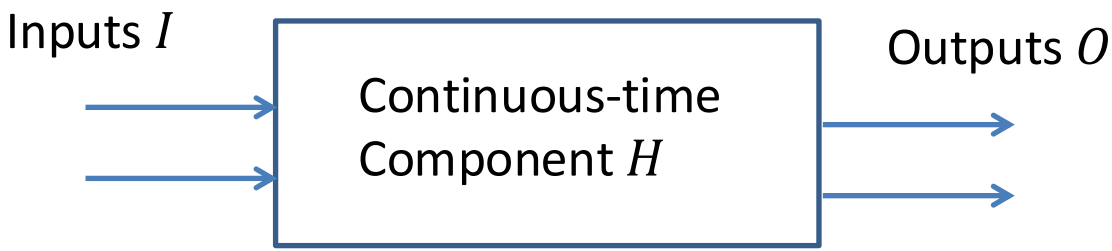
Stable, but not asymptotically stable

Equilibrium state 2: $\nu = 0$; $\varphi = -\pi$;
Pendulum is vertically upwards

Unstable !



Input-Output Stability



- A continuous-time component H maps input signals $\boldsymbol{I}(t)$ to output signals $\boldsymbol{O}(t)$

- Input-output stability: If we change the input signal slightly, the output signal should change only slightly
  - Formally: if $||\boldsymbol{I}(t)|| \leq \Delta_{\mathrm{I}}$, then $||\boldsymbol{O}(t)|| \leq \Delta_{\mathrm{O}}$

Euler and Runge Kutta

NUMERICAL METHODS FOR ODES

Algorithms for solving differential equations

- Exact formulation:

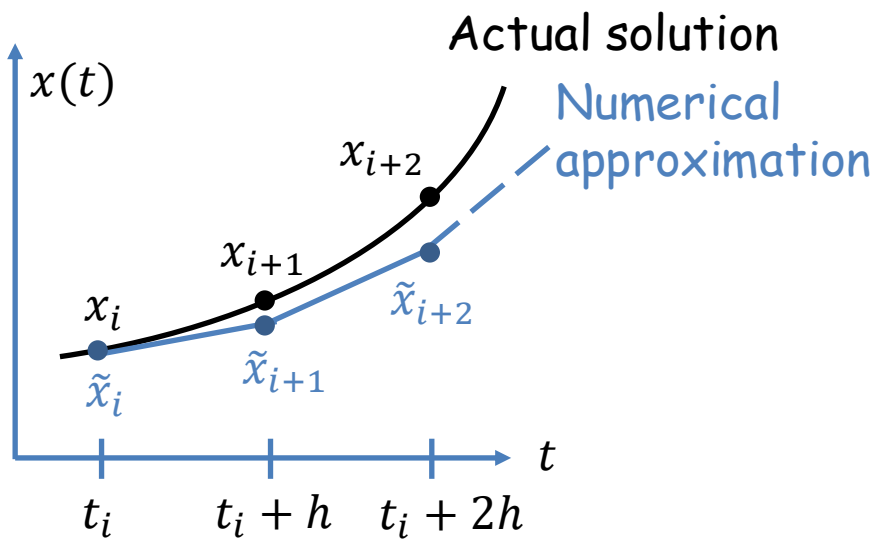
$$\frac{d}{dt}x(t) = f\big(x(t)\big)$$

- Approximation:

$$\frac{x(t+h) - x(t)}{h} \approx f\big(x(t)\big)$$

$$\Leftrightarrow$$

$$x(t+h) \approx x(t) + h \cdot f\big(x(t)\big)$$



- Euler method:

Input: f, x_0, h, N

$\tilde{x}_0 \leftarrow x_0$
for $i \in \{0,1,\dots,N\}$ do
    print $\tilde{x}_i$
    $\tilde{x}_{i+1} \leftarrow \tilde{x}_i + h \cdot f(\tilde{x}_i)$
end

- Theorem

$$\exists C > 0. \forall h > 0. \forall i \in \{0,1,\dots,N\}:$$

$$|\tilde{x}_i - x_i| \leq C \cdot h$$

So, the global truncation error is proportional to h.

Example of the Euler method

- Let $\frac{d}{dt}x = t$, $x_0 = 0$, $h = 1$, and $N = 5$
- The exact solution is $x(t) = \frac{1}{2}t^2$

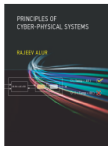
Iteration i	Exact solution x_i	Numerical solution $\tilde{x}_i$
0	0	0
1	0.5	$0 + 1 \cdot 0 = 0$
2	2	$0 + 1 \cdot 1 = 1$
3	4.5	$1 + 1 \cdot 2 = 3$
4	8	$3 + 1 \cdot 3 = 6$
5	12.5	$6 + 1 \cdot 4 = 10$

- Euler method:

Input: f, x_0, h, N

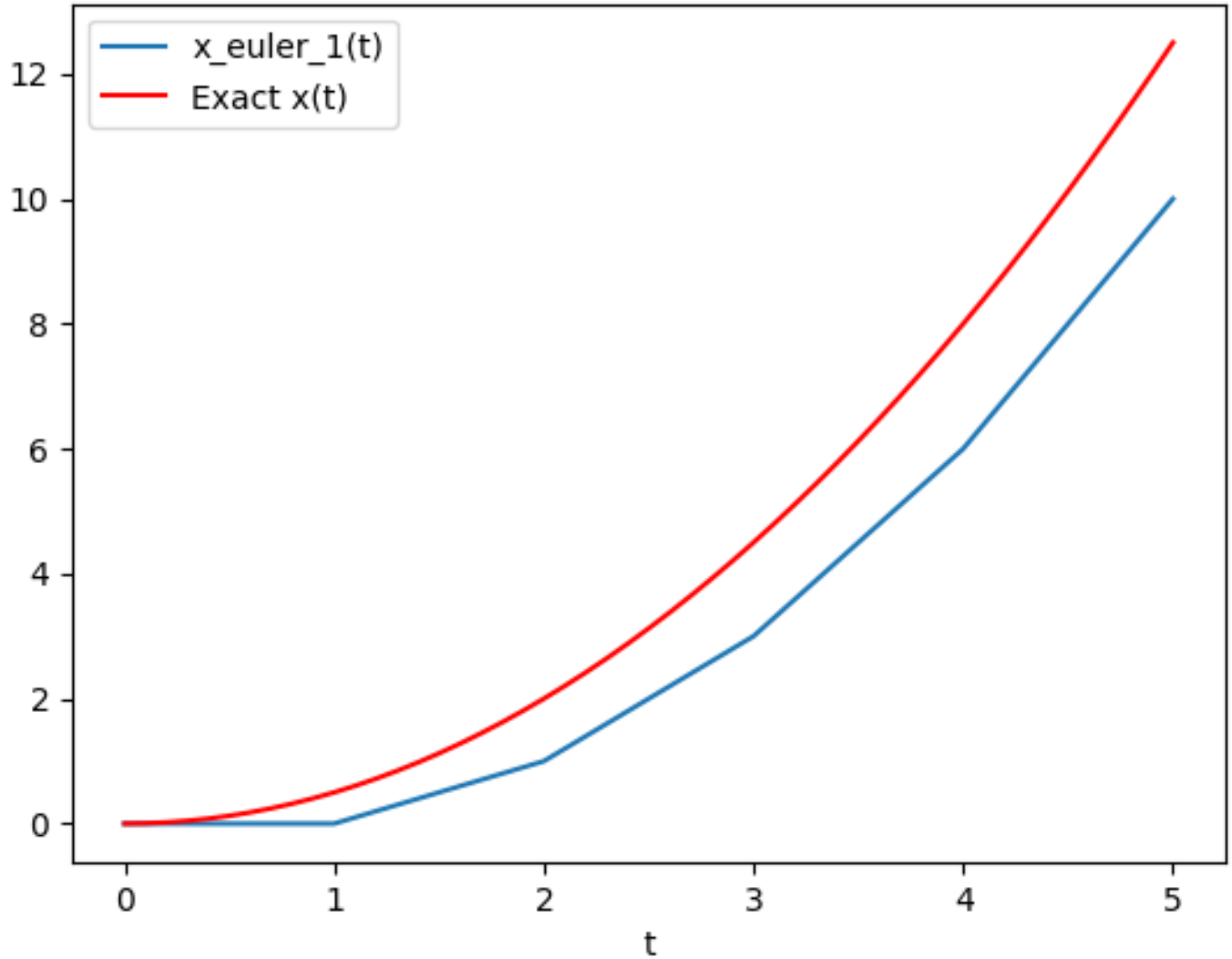
$\tilde{x}_0 \leftarrow x_0$
for $i \in \{0,1,\dots,N\}$ do
    print $\tilde{x}_i$
    $\tilde{x}_{i+1} \leftarrow \tilde{x}_i + h \cdot f(\tilde{x}_i)$
end

- Notice that $|\tilde{x}_i - x_i| = \frac{1}{2}i$



Precision of the Euler method

- Solution plotted with Euler method with different step sizes h



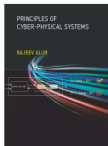
Algorithms for solving differential equations

- The Euler method assumes that the rate of change is constant during the interval, and is only based on the state at the beginning of the interval.
- This is often *not* the case for dynamical systems.
- We could take the endpoint into account as well.
- Runge-Kutta methods are based on this idea

- Second-order Runge-Kutta method uses the following calculations:

$k_1 = f\big(x(t)\big)$ ← Slope at beginning of interval
$k_2 = f(\tilde{x}_i + h \cdot k_1)$ ← Estimated slope at end of interval
$\tilde{x}_{i+1} = \tilde{x}_i + h \cdot (k_1 + k_2)/2$ ← Euler method, but now average slope

- Most commonly used method is the fourth-order Runge-Kutta method.



Algorithms for solving differential equations

- Most used method is the fourth-order Runge-Kutta method.
- It takes four slopes into account: start, two midpoints and endpoint.

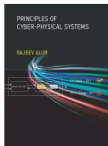
- $\tilde{x}_{i+1} = \tilde{x}_i + h(k_1 + 2k_2 + 2k_3 + k_4)/6$
- See book for expressions of k_1, k_2, k_3, k_4.

- Theorem

$$\exists C > 0. \forall h > 0. \forall i \in \{0,1, \dots, N\}:$$

$$|\tilde{x}_i - x_i| \leq C \cdot h^4$$

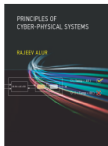
- No that for $0 < h < 1$: $h^4 \ll h$
- So, the fourth-order Runge-Kutta method is more efficient than the Euler method.



Continuous-time Component Exercise

From exercise sheet 10

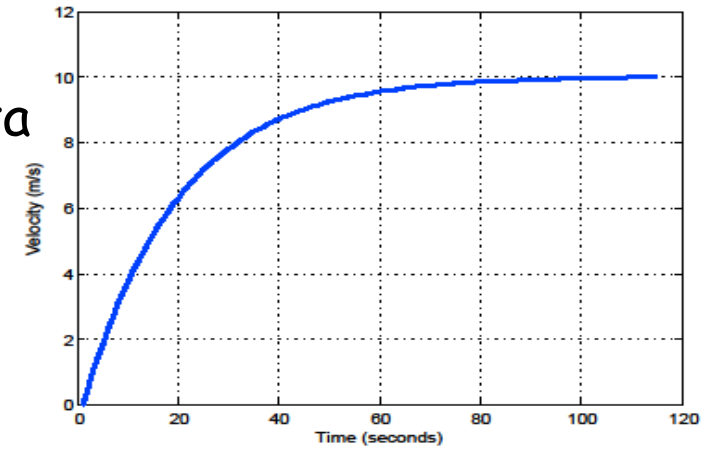
- Exercise 1 - implement Euler method – **<u>CRUCIAL, DO IT</u>**
- For the dedicated) Exercise 3 - implement the Runge-Kutta method



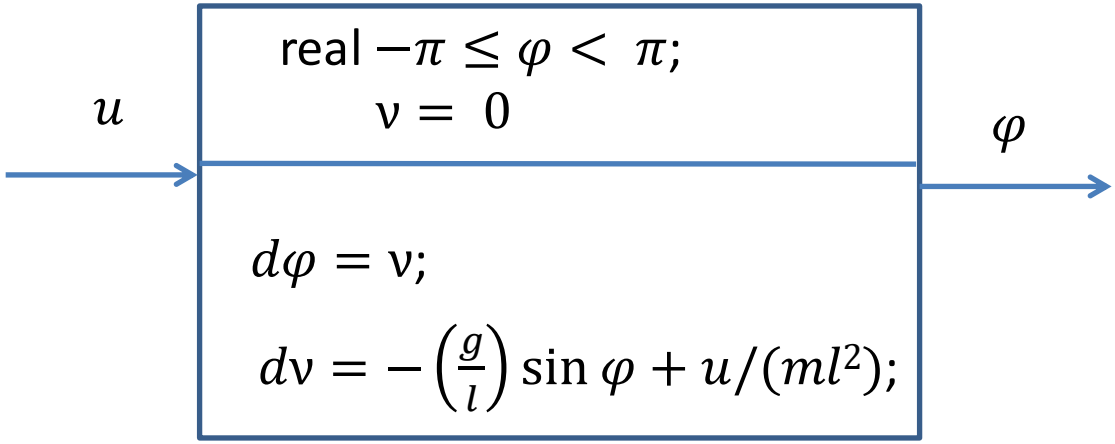
Back to Car Executions

- Suppose initial position is 0, initial velocity is 0, and force is constant F_0. Then, to get executions, we need to solve:
  - $\boldsymbol{x}(0) = 0;\ \boldsymbol{v}(0) = 0$
  - $d\boldsymbol{x}(t)/dt = \boldsymbol{v}(t)$
  - $d\boldsymbol{v}(t)/dt = (F_0 - k\,\boldsymbol{v}(t))\ /\ m$

- Compute the solution using Runge-Kutta
  - Mass $m = 1000kg$
  - Coefficient of friction $k = 50$
  - Force $F_0 = 500$ Newton
  - Velocity converges to $10\ m/s$



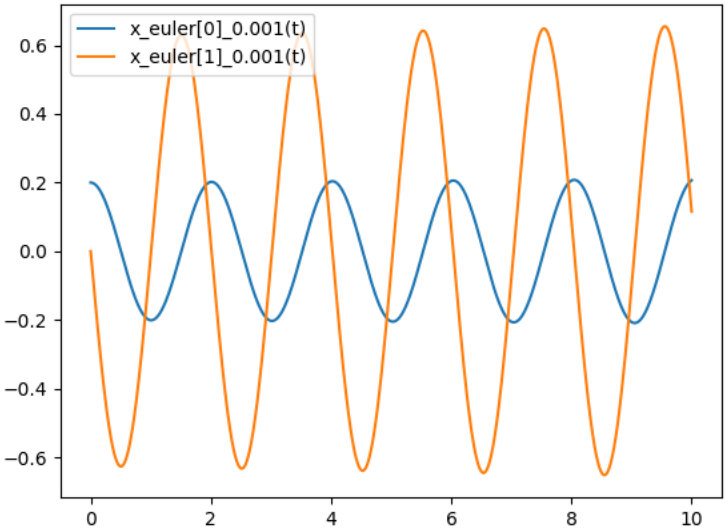
Simulation of the pendulum in Python




```
def f(x, t):
    # Implements the pendulum function without input
    g = 9.81
    l = 1
    result = [0, 0]

    result[0] = x[1]
    result[1] = -(g/l)*math.sin(x[0])

    return result
```



Equilibrium analysis with numerical methods

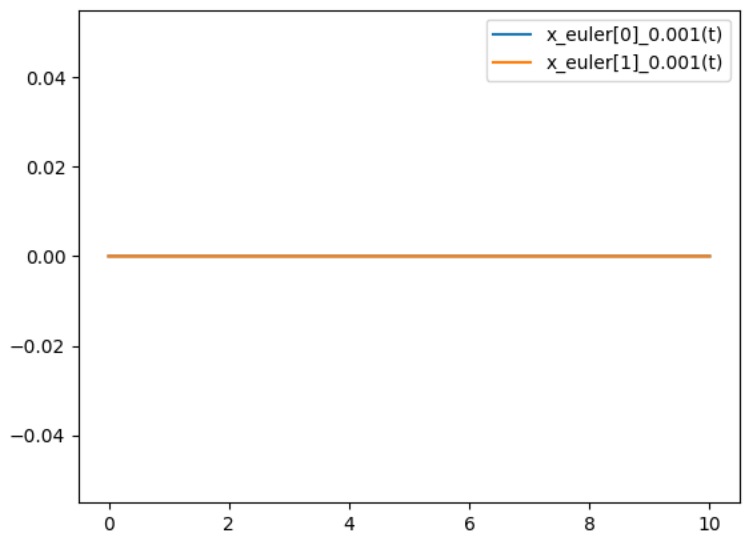
- Consider equilibrium state 1: $\varphi = 0; \nu = 0;$

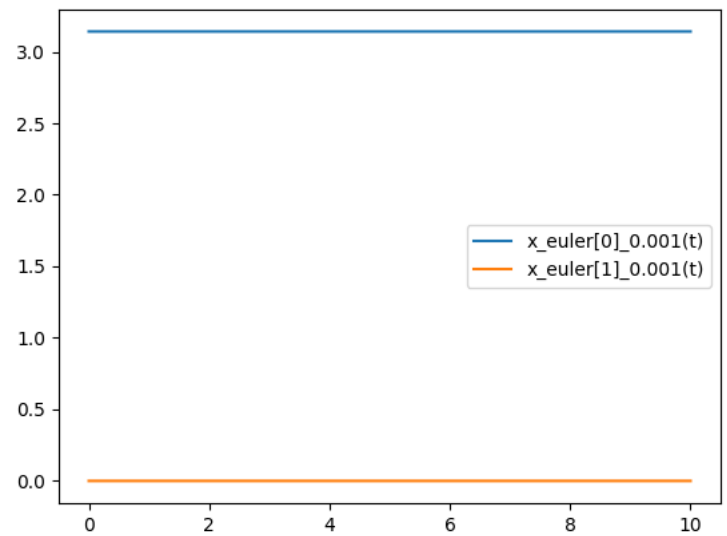
  In Python: x0 = [0, 0]

- Consider equilibrium state 2: $\varphi = -\pi; \nu = 0;$

  In Python: x0 = [3.14, 0]

  Better initial state: x0 = [math.pi, 0]





Instability

- If a continuous system is stable, it is considered to be correct.

- But what if a continuous system is unstable? Should we say to the mechanical engineers that they did a bad job at designing the dynamical system?

- Sometimes you can change the physical design to make it stable (towing a trailer example: https://youtu.be/6mW_gzdh6to?t=6), but this is not always possible.

- A Segway is an inverted pendulum (which is the unstable equilibrium point!)

- Solution: control software! (next lecture)

Dynamic Systems Exercise

From exercise sheet 8, do

- Exercise 5 – car model.
- Exercise 4 – Lipschitz continuity.